

1) Flashing of Led using AT89C51 Microcontroller using Proteus

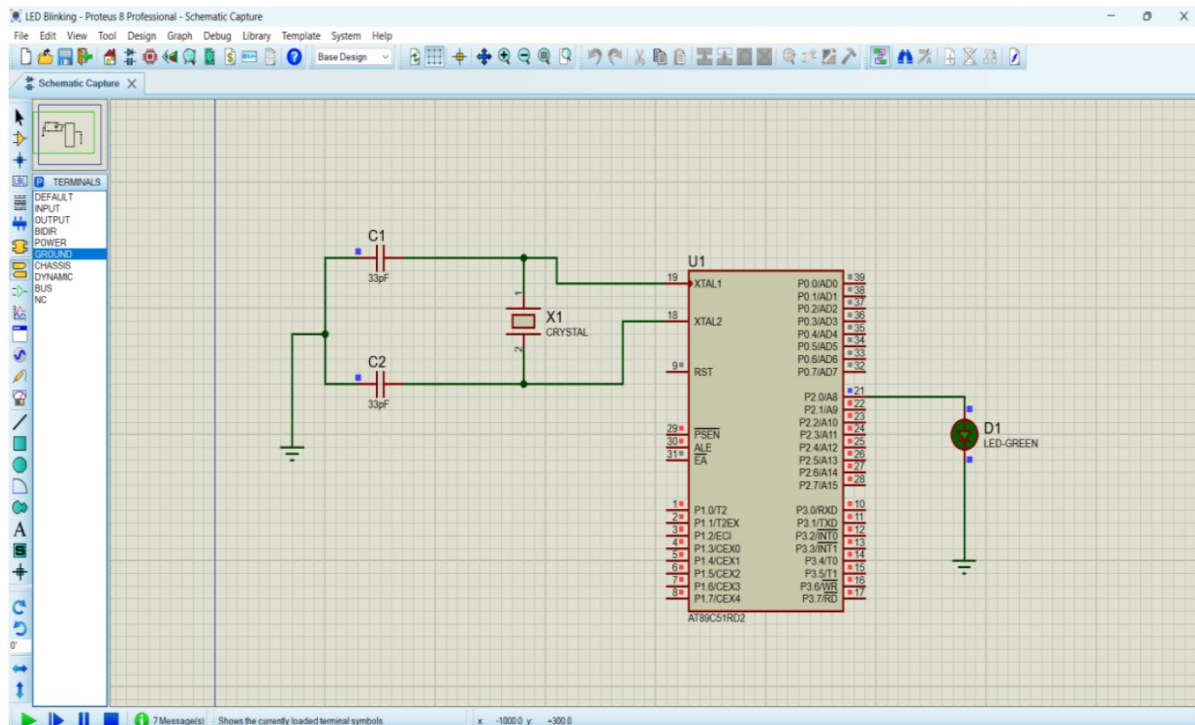
Objective:

- **Designed and simulated an LED blinking circuit using the AT89C51 microcontroller in Proteus.**
- **Developed an Embedded C program to turn the LED ON and OFF with a fixed time delay.**
- **Verified the correct LED blinking operation through Proteus simulation.**

Program:

```
ORG 0000H
UP: SETB P2.0
ACALL DELAY
CLR P2.0
ACALL DELAY
SJMP UP
DELAY: MOV R4,#35
H1:MOV R3,#255
H2:DJNZ R3,H2
DJNZ R4,H1
RET
END
```

Output:



2) GENERATION OF SQUARE WAVE USING PROTEUS

Objective:

- Designed and simulated a square wave generation circuit using Proteus.
- Generated a continuous square wave with the required frequency and duty cycle.
- Verified the square wave output using the virtual oscilloscope in Proteus.

Program:

ORG 0000H

UP: SETB P2.0

ACALL DELAY

CLR P2.0

ACALL DELAY

SJMP UP

DELAY: MOV R4,#35

H1:MOV R3,#255

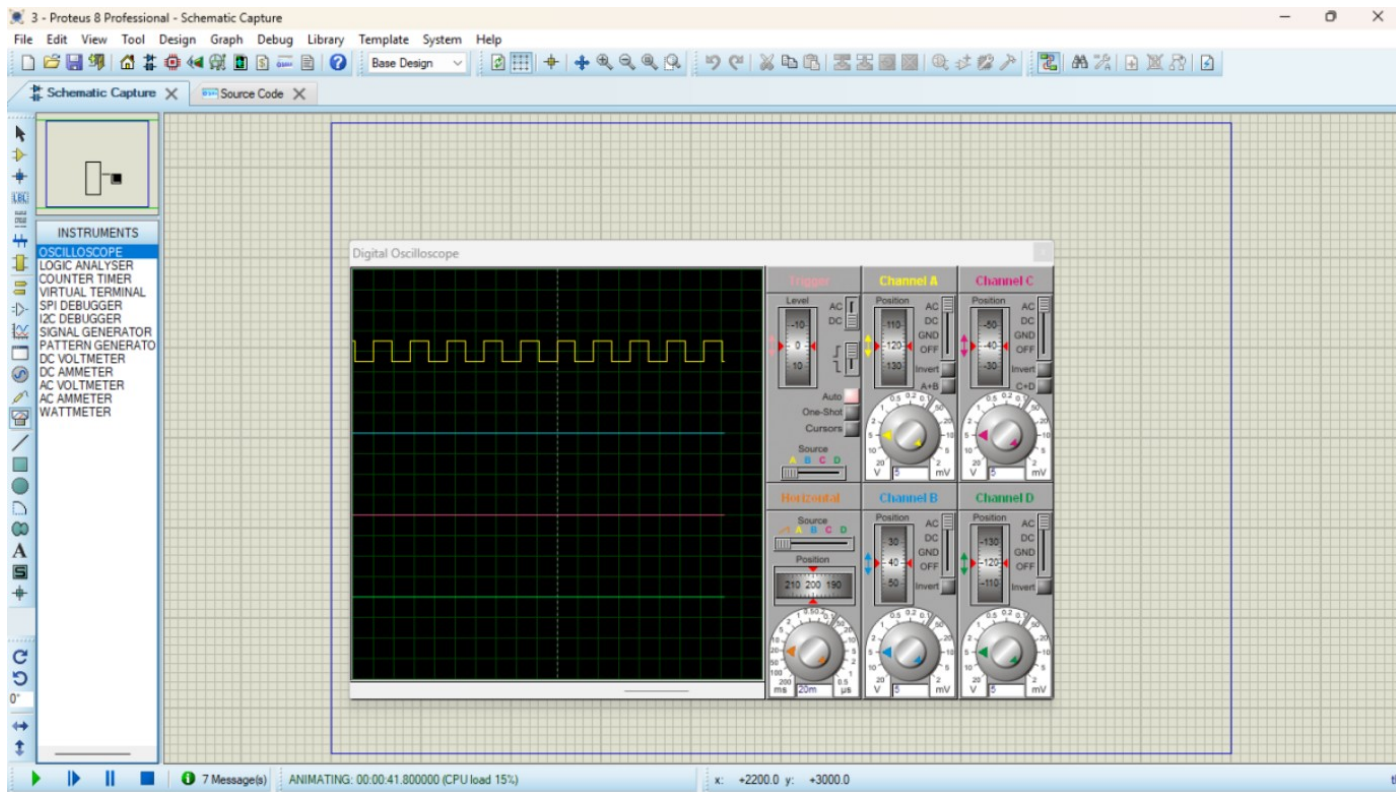
H2:DJNZ R3,H2

DJNZ R4,H1

RET

END

Output:



3) STEPPER MOTOR USING AT89C51 USING PROTEUS

Objective:

- Designed and simulated a stepper motor control circuit using the AT89C51 microcontroller in Proteus.
- Developed an Embedded C program to rotate the stepper motor in the desired sequence.
- Verified the correct rotational movement of the stepper motor through Proteus simulation.

Program:

```
ORG 0000H
```

```
UP: MOV P2,#09H
```

```
ACALL DELAY
```

```
MOV P2,#0CH
```

```
ACALL DELAY
```

```
MOV P2,#06H
```

```
ACALL DELAY
```

```
MOV P2,#03H
```

```
ACALL DELAY
```

```
SJMP UP
```

```
DELAY:MOV R4,#18
```

```
H1:MOV R3,#255
```

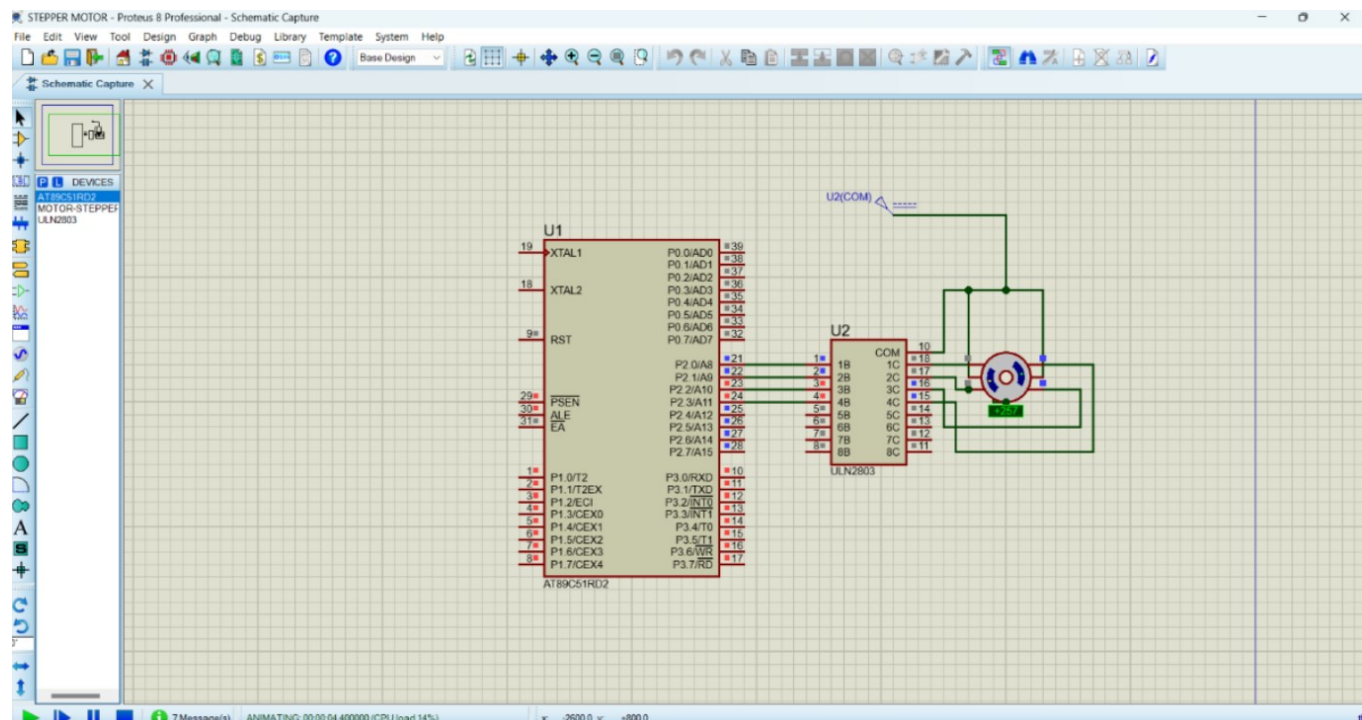
```
H2:DJNZ R3,H2
```

```
DJNZ R4,H1
```

```
RET
```

END

Output:



4) 7 SEGMENT DISPLAY USING AT89C51 USING PROTEUS

Objective:

- Designed and simulated a 7-segment display interface with the AT89C51 microcontroller in Proteus.
- Developed an Embedded C program to display numeric digits on the 7-segment display.
- Verified the correct display of numbers through Proteus simulation.

Program:

```
ORG 000H
UP:MOV P2,#0C0H
ACALL DELAY
MOV P2,#0F9H
ACALL DELAY
MOV P2,#0A4H
ACALL DELAY
MOV P2,#0B0H
ACALL DELAY
MOV P2,#99H
ACALL DELAY
MOV P2,#92H
ACALL DELAY
MOV P2,#82H
ACALL DELAY
MOV P2,#0F8H
ACALL DELAY
MOV P2, #80H
ACALL DELAY
MOV P2,#90H
```

```

ACALL DELAY
DELAY: MOV R5,#10
H1:MOV R4,#180
H2:MOV R3,#255
H3:DJNZ R3,H3
DJNZ R4,H2
DJNZ R5,H1
RET
END

```

Output:

